

Kafka Installation & Application - Final Notes

1. Update System

```
sudo apt-get update
```

2. Install SSH

```
sudo apt-get install openssh-client openssh-server  
sudo service ssh restart  
ssh localhost
```

3. Install Java

```
sudo apt update  
sudo apt install default-jdk  
java -version
```

4. Create User (Optional)

```
sudo adduser hadoop  
su - hadoop
```

5. Download Kafka

```
wget https://archive.apache.org/dist/kafka/3.2.1/kafka_2.12-3.2.1.tgz
```

6. Extract Kafka

```
tar -xzf kafka_2.12-3.2.1.tgz
```

7. Move & Create Symlink

```
sudo mv kafka_2.12-3.2.1 /home/  
sudo ln -s /home/kafka_2.12-3.2.1 /home/hadoop/kafka
```

8. Install Zookeeper

```
sudo apt-get install -y zookeeperd
```

9. Zookeeper Service File

```
[Unit]
```

```
Description=Apache Zookeeper server
```

```
After=network.target
```

```
[Service]
```

```
Type=simple
```

```
ExecStart=/home/hadoop/kafka/bin/zookeeper-server-start.sh /home/hadoop/kafka/config/zookeeper.properties
```

```
ExecStop=/home/hadoop/kafka/bin/zookeeper-server-stop.sh
```

```
Restart=on-abnormal
```

```
[Install]
```

```
WantedBy=multi-user.target
```

10. Kafka Service File

```
[Unit]
Description=Apache Kafka Server
Requires=zookeeper.service

[Service]
Type=simple
Environment="JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64"
ExecStart=/home/hadoop/kafka/bin/kafka-server-start.sh /home/hadoop/kafka/config/server.properties
ExecStop=/home/hadoop/kafka/bin/kafka-server-stop.sh

[Install]
WantedBy=multi-user.target
```

11. Start Services

```
sudo systemctl daemon-reload

sudo systemctl enable zookeeper
sudo systemctl start zookeeper
sudo systemctl status zookeeper

sudo systemctl enable kafka
sudo systemctl start kafka
sudo systemctl status kafka
```

12. Verify Kafka Files

```
ls /home/hadoop/kafka/bin
```

13. Create Topic

```
cd ~/kafka

bin/kafka-topics.sh --create \
--topic test-topic \
--bootstrap-server localhost:9092 \
--partitions 1 \
--replication-factor 1
```

14. Producer

```
bin/kafka-console-producer.sh --topic test-topic --bootstrap-server localhost:9092
```

15. Consumer

```
cd ~/kafka

bin/kafka-console-consumer.sh \
--topic test-topic \
--from-beginning \
--bootstrap-server localhost:9092
```

Final Flow

- 1 Install Java and SSH
- 2 Download and extract Kafka
- 3 Install Zookeeper
- 4 Configure services
- 5 Start services
- 6 Create topic
- 7 Use producer and consumer

Common Errors

- 1 404 error → use archive.apache.org
- 2 File not found → check ls
- 3 Permission denied → use sudo
- 4 Zookeeper error → fix path
- 5 Kafka not found → use find command

Viva Points

- 1 Kafka: Distributed streaming platform
- 2 Zookeeper: Coordination service
- 3 Topic: Message category
- 4 Producer: Sends data
- 5 Consumer: Reads data